Homework/Programming Assignment #2 **Homework/midterm Due: 03/24/2026- 3:30 PM**

Name/EID:
Email:
Signature (*required*)
I/We have followed the rules in completing this Assignment.

Name/EID:
Email:
Signature (*required*)
I/We have followed the rules in completing this Assignment.

Question	Points	Total
HA 1	25	
HA 2	25	
HA 3	25	
HA 4	25	
PA	100	
PA. k (Bonus)	15	
PA. m (Bonus)	30	
Presentation* (Bonus)	20	

Instruction:

1. Remember that this is a graded assignment. It is the equivalent of a **midterm take-home exam**.
2. *** You should present the results of the PA in the class** and receive extra bonus depending on the quality of your presentation!
3. **For PA questions**, you need to write a report showing how you derived your equations, describes your approach, test functions, and discusses the results. You should show your test results for each function.
4. You are to work **alone** or **in teams of two** and are **not to discuss the problems with anyone** other than the TAs or the instructor.
5. **You may use large language models (LLMs) such as ChatGPT or Gemini. If you do so, you must clearly state: *"I used ChatGPT to assist in responding to this assignment."* In addition, you must include the prompts you used and briefly describe how you validated the model's responses. The full sequence of prompts should be appended to your submission.**
6. It is open book, notes, and web. But you should cite any references you consult.
7. Unless I say otherwise in class, it is due before the start of class on the due date mentioned in the Assignment.
8. **Sign and append** this score sheet as the first sheet of your assignment.
9. Remember to submit your assignment in Canvas.

Homework Assignment (HA)

1. **Exercise 4.8** (pg.161) in *Modern Robotics: Mechanics, Planning, and Control* (Lynch et al.) [1].
2. **Exercise 4.11** (pg.162) in *Modern Robotics: Mechanics, Planning, and Control* (Lynch et al.) [1].
3. **Exercise 5.12 (**pg.208) in *Modern Robotics: Mechanics, Planning, and Control* (Lynch et al.) [1].
4. **Exercise 5.13** (pg.209) in *Modern Robotics: Mechanics, Planning, and Control* (Lynch et al.) [1].

➢ Programming Assignment (PA)

[Franka-Emika](#) [Kuka-Quantec](#) [Kuka LBR](#) [Yaskawa SDA10F](#)

Figure shows some of the robots that you have selected in the beginning of semester. For each of the selected robots, you first need to refer to the provided datasheet and documents by the company including their link dimensions and range of motions of each joints to answer the following questions. You need to completely cite the references that you are using to find these information:

- **a)** Find the forward kinematics (FK) of your selected robot using the **space form of the exponential products.**
- **b)** Write the function **"FK_space.m"** that calculates the *space form FK* of the robot and represents the defined frames and screw axis graphically. **Note**: your program should be modular and generic such that it can be used for any type of serial open-chain manipulator!
- **c)** Repeat (a) and (b) for the **body form FK** and write a function **"FK_body.m"**.
- **d)** Find the **space and body form** Jacobian of the robot.
- **e)** Write functions **"J_space.m"** and **"J_body.m"** that calculate the **space and body-form Jacobians** of robot, respectively.
- **f)** Write a function **"singularity.m"** that analytically calculates the **singularity configurations** of the robot**.**
- **g)** Write functions that based on the Jacobian of the robot at each configuration:
    - a. show/plot the manipulability ellipsoids for the linear and angular velocities and their axes (i.e., **"ellipsoid_plot_angular.m" and "ellipsoid_plot_linear.m").**
    - b. Calculate the isotropy **"J_isotropy.m"**, condition number **"J_condition.m"**, and volume of the ellipsoids **"J_ellipsoid_volume.m"**.
- **h)** Using the derived forward kinematics and Jacobians, write a function **"J_inverse_kinematics.m"**, that uses the iterative numerical inverse kinematics algorithm to control the robot from arbitrary **configuration a** to desired **configuration b**. Test your algorithm in various configurations.

- **i)** Use **Jacobian Transpose algorithm** (**“J_transpose_kinematics.m”**), and repeat **part h**.
- **j)** Using the redundancy resolution approach that we discussed in the class, extend your function in **part h** to define a secondary objective function of the joint variables that **maximizing this manipulability measure and** exploits redundancy to move away from singularities (**“redundancy_resolution.m”**).
- **k)** (**Bonus**) extend your written function **(“DLS_inverse_kinematics.m”)** in **part h** such that utilizing the **Damped Least Square Approach** can control the robot at configurations **near the singularity situations.** **Hint:** you may use your written functions in g) and h) to detect singularity situations and switch to this mode for controlling the robot.
- **l)** **For each function**, you should write test functions and discuss the results you obtained in your report. **Do not hand in** only functions, you will receive half of the points if you do not hand in appropriate reports and modular functions!
- **m)** (**Bonus**) Graphical simulation of the robot in ROS, Matlab Robotic System Toolbox, VREP, or similar software has extra bonus.

➢ References:

1. Lynch and Park, “***Modern Robotics***,” Cambridge U. Press, 2017, **Chapter 3.**
2. Bruno, Siciliano, Sciavicco Lorenzo, Villani Luigi, and Oriolo Giuseppe. "*Robotics: modelling, planning and control*." Advanced Textbooks in Control & Signal Processing 4 (2009): 76-82, **Chapter 3**.